Exercise 1 — PyMuPDF

Install:

```
python -m pip install pymupdf
```

Use `EY_Digital_PDF_Practice.pdf`.

Tasks:

1. Open the PDF.
2. Print the number of pages.
3. Extract and print text from page 1.
4. Extract text from all pages.
5. Search whether the word `maintenance` appears.
6. Count how many times `V303` appears.
7. Extract only page 2 text.
8. Save all extracted text into `digital_pdf_text.txt`.

Starter:

```
import fitz

doc = fitz.open("EY_Digital_PDF_Practice.pdf")

print("Pages:", len(doc))

for page in doc:
    print(page.get_text())
```

Exercise 2 — PyPDF

Install:

```
python -m pip install pypdf
```

Use the same digital PDF.

Tasks:

1. Open the PDF using `PdfReader`.
2. Print the number of pages.
3. Extract text from the first page.
4. Extract text from the second page.
5. Combine text from both pages into one string.
6. Check whether `NorthStar Maintenance` exists in the PDF.
7. Find the number of occurrences of the word `vendor`.
8. Save the extracted content into `pypdf_output.txt`.

Starter:

```
from pypdf import PdfReader

reader = PdfReader("EY_Digital_PDF_Practice.pdf")

print("Pages:", len(reader.pages))

for page in reader.pages:
    print(page.extract_text())
```

Exercise 3 — Tesseract OCR

Use `EY_Scanned_PDF_OCR_Practice.pdf`.

Install:

```
python -m pip install pymupdf pytesseract pillow
```

Tesseract itself must also be installed on the machine.

Tasks:

1. Open the scanned PDF using PyMuPDF.
2. Convert the PDF page into an image.
3. Run Tesseract OCR on the image.
4. Print the extracted text.
5. Check whether `Forklift FL-17` was detected.
6. Search for the word `injuries`.
7. Extract lines mentioning `inspection`.

8. Save the OCR output into `ocr_result.txt` .
9. Count the number of times `inspection` appears.
10. Print all lines containing `Loading Bay` .

Example:

```
import fitz
import pytesseract
from PIL import Image

doc = fitz.open("EY_Scanned_PDF_OCR_Practice.pdf")

page = doc[0]

pix = page.get_pixmap()

pix.save("scanned_page.png")

image = Image.open("scanned_page.png")

text = pytesseract.image_to_string(image)

print(text)
```